

PicoCTF 2026 – StegoRSA

StegoRSA

Cryptography

Easy

100 pts 5,984 solves

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A message has been encrypted using RSA. The public key is gone... but someone might have been careless with the private key. Can you recover it and decrypt the message?

Download the [flag](#) and [image](#).

In this challenge, they want us to decrypt an RSA-encrypted message using an RSA private key

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Exif Data Extraction

```
Comment          : 2d2d2d2d2d424547494e205052  
4956415445204b45592d2d2d2d2d0a4d494945766749424144414e42676b  
71686b6947397730424151454641415343424b67776767536b4167454141  
6f49424151445942674148797171517a4f58430a756c364f752f4a6e4443
```

The first thing we notice about this challenge is that the picture file that's part of the challenge has some strange content in its EXIF metadata

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Hex to ASCII Encoding

```
Comment : 2d2d2d2d2d424547494e205052  
4956415445204b45592d2d2d2d2d0a4d494945766749424144414e42676b  
71686b6947397730424151454641415343424b67776767536b4167454141  
6f49424151445942674148797171517a4f58430a756c364f752f4a6e4443
```

We suspect that this is hexadecimal encoding of ASCII characters, and we can use the xxd program to get the original content back

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RSA Private Key for Decryption

```
-----BEGIN PRIVATE KEY-----  
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBAKgwggSkAgEAAoIBAQDYBgAHyqqQzOXC  
u160u/JnDC54T5xRwn+kN38MmmF9o+OK4048zIFfIBW+020z7rGm8FKQUo1m1MY/  
POXjTz0/KKYKPbuJ322IWM1AAvBmkm1zRu6D2Td9joRZ1dIpEDCe5jru9IIF2G6K
```

The original content is an RSA private key, and we can use a private key to decode an RSA-encrypted message

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RSA Decryption with Private Key

```
theshyhat-academy@webshell:~/stego_rsa$ openssl pkeyutl -decrypt -inkey key  
-in flag.enc -out flag.txt  
theshyhat-academy@webshell:~/stego_rsa$ cat flag.txt  
picoCTF{rsa_key_in_img_20588810}
```

After decrypting the encrypted message with the private key, we can read the flag